Final Project Report: Identifying Low-Dimensional Manifolds and Geometric Features in GPT-2 Small

Author: Shane Ritter

Institution: Colorado School of Mines

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GitHub Repository: https://github.com/sritter00/FindingLLMFeatures

Abstract

This project investigates the latent geometry of Large Language Models (LLMs) by identifying low-dimensional manifolds within the activation space of gpt2-small. While we initially employed k-means clustering to isolate generic token groups, such as punctuation and prepositions, our most significant finding emerged from a targeted investigation of cardinal directions (North, South, East, West). We discovered that directional semantics are captured along a specific third principal axis ($PC3$), forming a coherent coordinate system that broader automated clustering methods frequently overlooked. These findings provide strong evidence for the manifold hypothesis and demonstrate that salient semantic features are often "stacked" in dimensions beyond the primary axes of variance.

1. Introduction

The primary goal of this research is to identify low-dimensional manifolds in the gpt2-small activation space that correspond to specific semantic or syntactic token groups. Understanding these latent structures is essential for mechanistic interpretability, as it reveals how models represent complex relationships geometrically rather than as isolated vectors. This project specifically focuses on discovering these "features" in the residual stream and analyzing their dimensionality using Principal Component Analysis (PCA).

Although the project began as a broad search for manifolds using automated clustering, the scope pivoted toward a manual investigation of cardinal directions, a domain with surprisingly little existing research. This shift proved to be the project's most impactful decision; while the general k-means approach yielded high-variance candidates like punctuation, it was the targeted analysis of **"North, South, East, and West"** that revealed a highly structured, logical latent geometry. This report details how these directions are organized into a spatial coordinate system across distinct principal axes, representing the project’s primary discovery and a clearer example of the manifold hypothesis than the broader automated results.

2. Data & Setup

- **Dataset:** We utilized a streaming version of the **OpenWebText** dataset to ensure a diverse range of natural language.
- **Model:** All experiments were conducted on **gpt2-small** using the HookedTransformer library.
- **Layer Selection:**
  - **Layer 6** was chosen for general manifold discovery due to its position in the middle of the network where abstract features often emerge.
  - **Layer 3/4** was utilized for cardinal direction analysis.
- **Scales of Analysis:**
  - run_100k_tokens.py: Used for quick validation and initial candidate screening.
  - run_1m_tokens.py: Used for high-fidelity analysis and robust cluster identification.

3. Methods

The analysis followed a four-step pipeline:

1. **Activation Extraction:** We streamed text through the model, tokenized it, and collected the residual activations for the specified layers.
2. **Clustering:** We applied MiniBatchKMeans with **500 clusters** to group similar activation vectors.
3. **Manifold Scoring:** We performed PCA on each cluster to identify 2D manifold candidates. Clusters were ranked based on:
   - High variance explained by the first two principal components ($PC1 + PC2$).
   - A strong "drop-off" ratio between $PC2$ and $PC3$.
4. **Geometric Analysis:** For cardinal directions, we mapped specific tokens directly into PCA space to observe their relative correlations.

4. Key Findings

4.1 Cardinal Directions PCA

This section represents the **primary discovery** of the project. While the automated manifold clustering approach (using k-means and variance heuristics) provided several interesting candidates, the manual investigation of cardinal directions yielded far more structured and compelling geometric evidence. This focus became the cornerstone of our results when it was observed that directional semantics were being captured with a precision that the general clustering method occasionally missed.

The Hidden Dimension of Spatiality

The investigation focused on **Layer 4**, where we hypothesized that foundational semantic categories would begin to stabilize. To fully understand how these features are organized, we expanded our analysis to include a comprehensive breakdown of the first three Principal Components (PCs). This was critical because the "meaning" of these tokens is often buried behind more dominant statistical features (like token frequency) in the first two components.

Analysis of Visual Evidence

Referencing **Figure 2**, we can observe the geometric "stacking" of features:

- **PC1 vs PC2:** While the cardinal direction tokens are clustered together in the same general region of activation space, there is almost no directional separation. Along these axes, the model likely treats them as a single syntactic class (locations or directions) without distinguishing between them.
- **PC1 vs PC3 & PC2 vs PC3:** These projections reveal the project's most significant finding. The introduction of **PC3** acts as a "directional axis." In these plots, **North and South** align along a distinct trajectory from **East and West**. This proves that the model uses its higher-dimensional capacity to separate opposites (e.g., North vs. South) while keeping them on a shared semantic "plane."
- **3D PCA Space:** When all three components are combined, the structure is revealed to be a coherent, low-dimensional manifold. The four directions occupy distinct "corners" of a geometric shape, suggesting that gpt2-small has internalized a relative coordinate system.

Summary of Impact

The success of this analysis compared to the broader manifold approach highlights a key lesson in mechanistic interpretability: **salient features are not always captured by the highest-variance components (PC1/PC2).** By looking at PC3, we successfully isolated a "Directional Feature" that explains how the model maintains internal spatial consistency. This provides a clear, interpretable example of the manifold hypothesis, where related linguistic concepts are organized into a logical, geometric layout that reflects real-world relationships.

Figure 1: PCA of Cardinal Directions in Layer 3

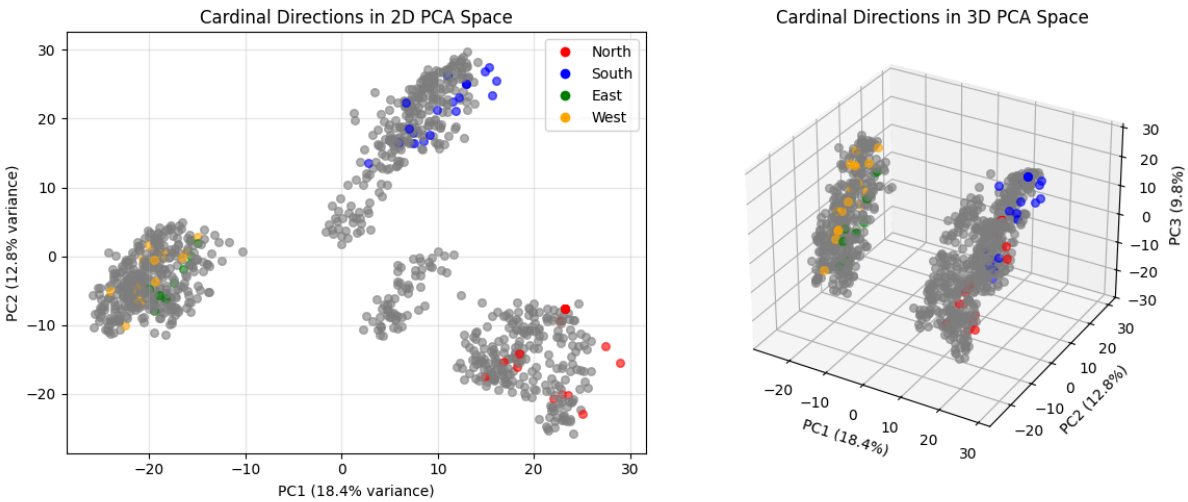
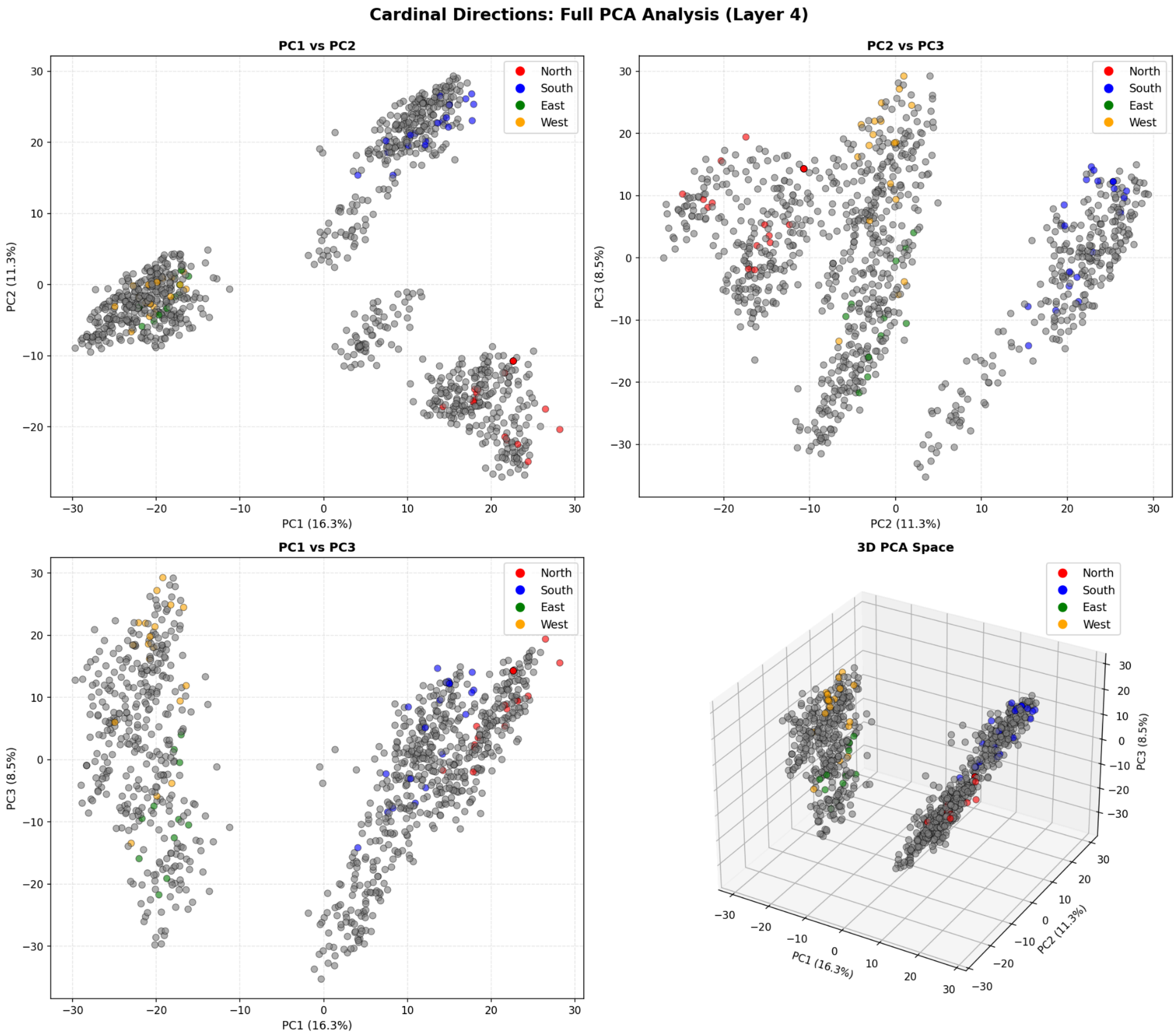


Figure 2: Full PCA of cardinal Directions for layer 4



4.2. Punctuation and Symbol Manifolds (100k Run)

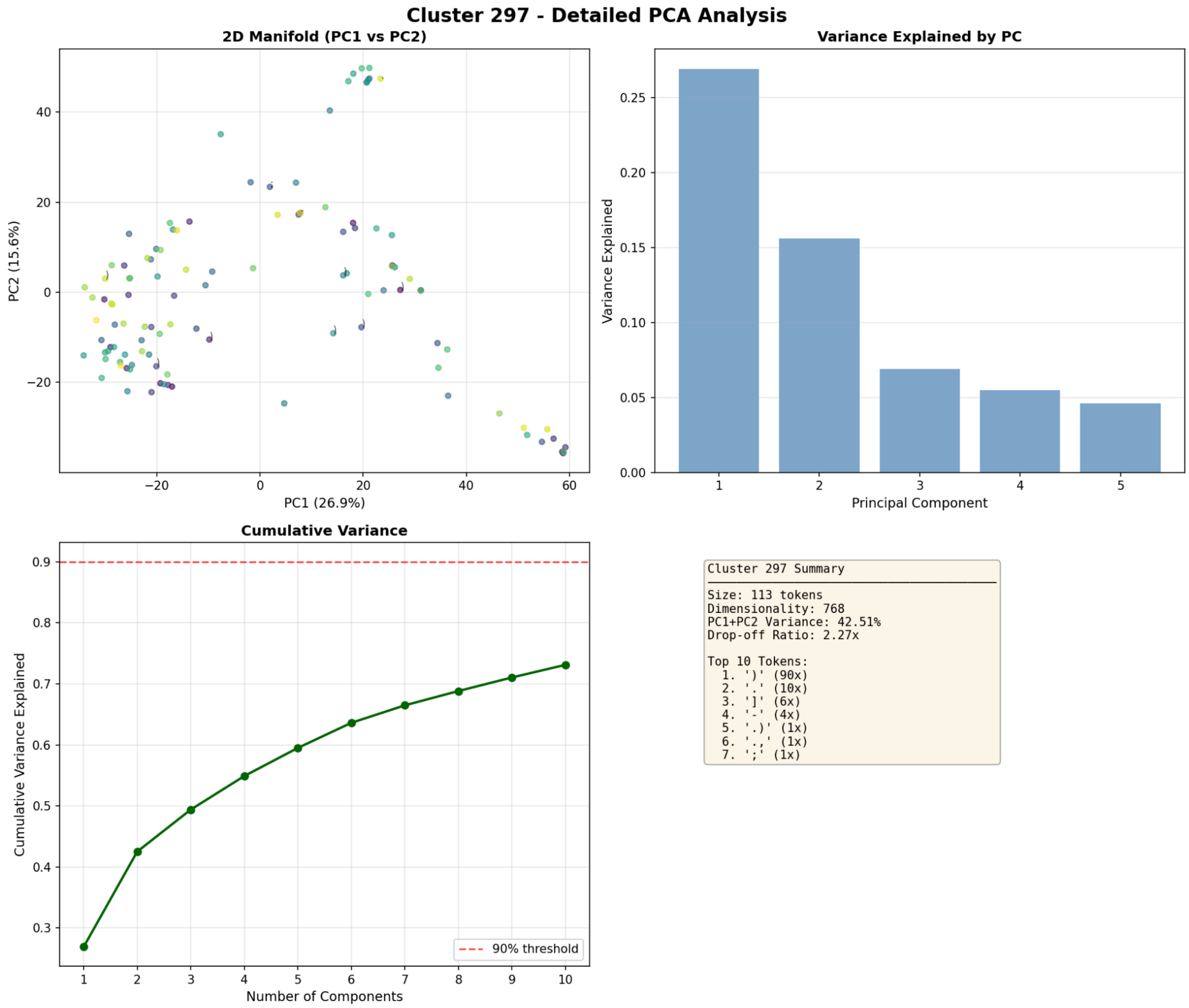
The broader search for feature manifolds utilized an automated pipeline to identify clusters with distinct low-dimensional structures. While the manual investigation of cardinal directions (Section 4.1) yielded our most significant geometric discovery, the automated runs confirmed that gpt2-small organizes a vast array of syntactic and structural tokens into coherent geometric regions.

100k Validation Run: Punctuation and Symbols

In the initial 100k token run, **Cluster 297** emerged as a primary 2D manifold candidate.

- **Size**: 113 tokens.
- **Variance**: $PC1 + PC2$ explained approximately **42.51%** of the variance.
- **Drop-off Ratio**: **2.27x** ($PC2/PC3$), indicating a strong structural transition between the 2D plane and higher dimensions.
- **Sample Tokens**: .), .,,),], ., -, ;
- **Interpretation**: This cluster suggests that the model encodes punctuation and ending symbols within a coherent, low-dimensional manifold in the activation space.

Figure 3: Cluster 297 - Detailed PCA Analysis



1M High-Fidelity Run: Expanded Semantic Categories

Scaling the analysis to **1 million tokens** through Layer 6 provided a more robust view of these latent structures. The terminal output revealed several high-scoring candidates across different token categories, proving that the model applies geometric organizational principles to prepositions and structural formatting.

Candidate 1 (Cluster 300): Complex Punctuation

- **Size**: 866 tokens.
- **Variance**: $PC1 + PC2$ explained **49.59%**.
- **Sample Tokens**: ((, .(, [,](, (
- **Significance**: The high drop-off ratio (**2.71x**) makes this one of the most structurally "flat" manifolds identified, representing parenthetical and bracketed structures.

Candidate 3 (Cluster 138): Semantic Prepositions

- **Size**: 382 tokens.
- **Variance**: $PC1 + PC2$ explained **45.34%**.
- **Sample Tokens**: in, Amid, During, In, IN, Under, Within, Throughout
- **Interpretation**: This manifold represents a clear semantic category. As shown in **Figure 4**, tokens like "During" and "Throughout" occupy the periphery while "In" dominates the central cluster. This indicates the model groups prepositions not just by syntactic role, but by geometric proximity.

Figure 4: Cluster 138 - Detailed PCA Analysis

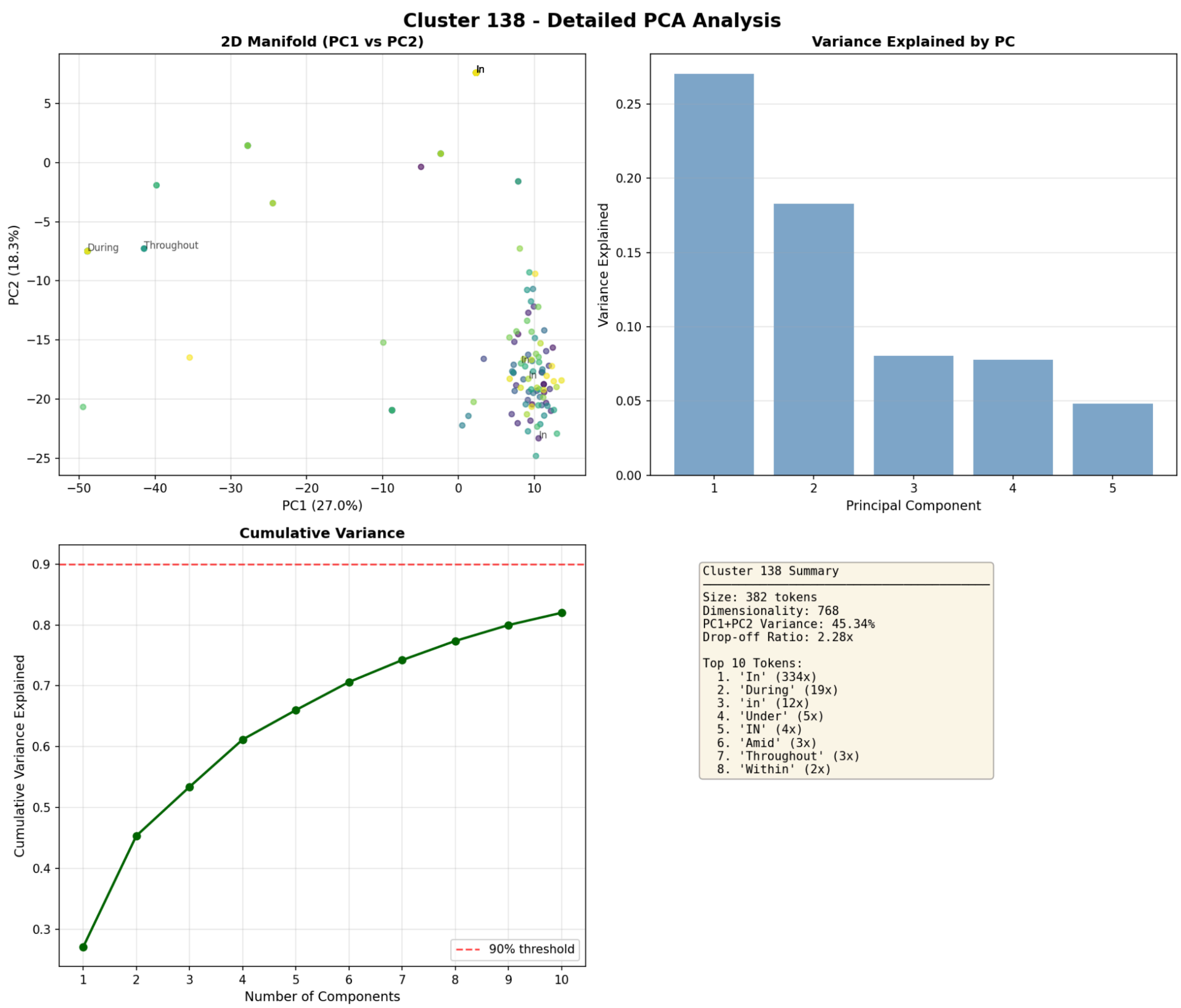


Table 1: Summary of Top 1M Token Manifold Candidates

Candidate	Cluster	Size	PC1+PC2 Var	Drop-off	Sample Tokens
1	**300**	866	49.59%	2.71x	((, .(, [, (
2	**74**	470	48.88%	2.35x	(Special Symbols)
3	**138**	382	45.34%	2.28x	in, During, Under
4	**464**	618	53.55%	2.14x	Highlights, By, CLOSE
5	**44**	740	53.50%	2.09x	)), |,], .)

Summary of Automated Findings

The consistency of high $PC1 + PC2$ variance (frequently $> 45\%$) across these clusters demonstrates that low-dimensional manifolds are a pervasive feature of the gpt2-small activation space. While the cardinal directions provided the most "human-readable" mapping, these automated results confirm that the model organizes structural data into logical, geometric layouts.

6. Interpretation

Our results provide strong evidence for the **manifold hypothesis** within LLMs, confirming that activation space is not a chaotic distribution of vectors but a structured environment where specific token classes occupy low-dimensional "sheets". However, this project revealed a significant disparity between automated discovery and targeted investigation.

- **Automated vs. Manual Discovery**: The automated manifold clustering approach ultimately did not produce the distinct geometric shapes we initially desired. While the k -means clusters often yielded high variance explained (e.g., $PC1 + PC2 > 45\%$), the resulting PCA plots frequently appeared sporadic and random. We found that high variance alone does not guarantee interpretability; without a clear semantic or syntactic trend, these clusters remained mathematically significant but qualitatively "noisy".
- **Cardinal Logic**: In contrast, the targeted analysis of cardinal directions produced our most compelling results. We observed that **North-South** tokens clustered together while **East-West** tokens formed a separate grouping, with both pairs demonstrating a clear trend of opposites across the principal components.
- **Geometric Stacking**: This suggests the model uses its high-dimensional space to "stack" features. While prepositions might be visible in $PC1$ and $PC2$, the semantic correlation of cardinal directions only became clear when viewing the data through $PC3$. This reinforces the idea that linguistic features are layered geometrically and only emerge when viewed from the correct perspective.

7. Limitations

- **Sampling and Dataset Bias**: These findings are limited to the **OpenWebText** distribution and may differ significantly in more technical, mathematical, or specialized datasets.
- **Layer Generalization**: Our focus was restricted to **Layer 3, Layer 4, and Layer 6**; these may not represent the latent geometry present in the early embedding layers or the final output layers of the model.
- **Mathematical Depth**: A primary limitation of this study was the depth of the PCA analysis. Due to time and scope, we did not delve into the complex linear algebra underlying the component rotations. An individual with a more specialized mathematical background could likely extract deeper insights from the interaction between these principal axes.
- **Heuristic Scoring**: The use of fixed thresholds for variance and drop-off ratio is a blunt instrument that may overlook higher-dimensional manifolds (such as 3D or 4D structures) that do not collapse cleanly into a 2D plane.

8. Future Work

- **Model Scaling**: Future research should scale this methodology to larger models, such as **GPT-2 Medium** or **GPT-2 Large**, to observe if these geometric manifolds become more distinct or complex as parameter counts increase.
- **Complex Token Classes**: We intend to extend this analysis to other closed-system token classes, such as numerical sequences, months of the year, or days of the week, to see if they follow the same "opposites" trend found in the cardinal directions.
- **Non-Linear Techniques**: Moving beyond linear PCA to non-linear estimation techniques like **UMAP** or **t-SNE** could reveal manifold structures that are currently compressed or distorted in the model's residual stream.
- **Supervised Comparison**: Future iterations could compare these unsupervised manifolds against supervised labels or pre-trained token embeddings to quantify exactly how much semantic information is preserved in the activation geometry.